Tutorial-8

Total entropy :- $H(S) = -P_S \log_2 P_S - P_N \log_2 P_N$

$$= -\frac{5}{10}\log_2 5/10 - \frac{5}{10}\log_2 5/10$$

$$= -0.5 \log_2 2^{-1} - 0.5 \log_2 2^{-1}$$

$$= 1$$

Information Gain (of features towards class) $= H(S) - H(S/A)$

$$= H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$$

⇒ for money keywords

Entropy $m=1 \Rightarrow -\frac{5}{5}\cancel{\log_2 5/5}^{0} - \frac{0}{5}\cancel{\log_2 0/5}^{0}$

$$= 0$$

$m=0 \Rightarrow -\frac{0}{5}\cancel{\log_2 0/5}^{0} - \frac{5}{5}\cancel{\log_2 0/5}^{0}$

$$= 0$$

$$H(S/A) = 1 - \frac{5}{10}\times 0 - \frac{5}{10}\times 0 = 1$$

⇒ Senders known,

⇒ 1	$\frac{5}{1}$	$\frac{10}{5}$
⇒ 0	4	0

$\Rightarrow$ Entropy $m=1 \Rightarrow \frac{-1}{6}\log_2 1/6 - \frac{5}{6}\log_2 5/6$

$= 0.65$

$m=0 \Rightarrow -\frac{4}{4}\log_2 4/4 - \frac{0}{4}\log_2 4/4$

$= 0$

$\Rightarrow H(S) - H(S/A) = 0.61$

$\Rightarrow$ Has Attachment,

	S	N
$\Rightarrow 1$	3	2
$\Rightarrow 0$	2	3

$\Rightarrow$ Entropy $m=0 \Rightarrow -\frac{2}{5}\log_2 2/5 - \frac{3}{5}\log_2 3/5$

$= 0.971$

$m=1 \Rightarrow -\frac{3}{5}\log_2 3/5 - \frac{2}{5}\log_2 2/5$

$= 0.971$

$\Rightarrow H(S) - H(S/A) = 1 - \frac{5}{10} \cdot 0.971 - 0.971\left(\frac{5}{10}\right)$

$= 0.029$

Data $< 4 \Rightarrow 0$

Data $\geq 4 \Rightarrow 1$

	S	N
$\Rightarrow 1$	5	0
$\Rightarrow 0$	0	5

$\Rightarrow$ Entropy $m=1 \Rightarrow -\frac{5}{5}\log_2 5/5 - \frac{0}{5}\log 0/5$

$= 0$

$m=0 \Rightarrow \frac{-0}{5}\log_2 0/5 - \frac{5}{5}\log 5/5$

$= 0$

$H(S) - H(S/A) = 1 - \frac{5}{10}\times 0 - \frac{5}{10}\times 0$

$= 1$

$\Rightarrow$ Suspicious words,

1
0
1
0
1
0
1
0
1
0